

Otto Intelligence Service - Complete API Documentation

Base URL: `https://ottoai.shunyalabs.ai`

Authentication: All endpoints (except `/health` and `/`) require an API key in the request header:

```
X-API-Key: 5q3fw1iU9ZFo3epTCsUfUiDw1Dy4DnBP
```

1. Health & Status APIs

1.1 Health Check

Endpoint: `GET /health`

Description: Check if the service is healthy and running.

Authentication: Not required

cURL:

```
curl -X GET https://ottoai.shunyalabs.ai/health
```

Response (200 OK):

```
{
  "status": "healthy",
  "service": "otto-intelligence",
  "version": "1.0.0",
  "environment": "production"
}
```

1.2 Root Information

Endpoint: `GET /`

Description: Get basic service information and links.

Authentication: Not required

cURL:

```
curl -X GET https://ottoai.shunyalabs.ai/
```

Response (200 OK):

```
{
  "service": "Otto Intelligence Service",
  "version": "1.0.0",
  "docs": "/docs",
  "health": "/health"
}
```

1.3 API Status

Endpoint: `GET /api/v1/status`

Description: Get API version and feature status.

Authentication: Required

cURL:

```
curl -X GET https://ottoai.shunyalabs.ai/api/v1/status \
-H "X-API-Key: your_api_key_here"
```

Response (200 OK):

```
{
  "api_version": "v1",
  "features": {
    "call_processing": "pending_implementation",
    "insights_engine": "pending_implementation",
    "ask_otto": "pending_implementation"
  },
  "database": {
    "mongodb": "connected",
    "redis": "connected",
    "milvus": "connected"
  }
}
```

1.4 Scheduler Status

Endpoint: GET /api/v1/scheduler/status

Description: Get the status of background scheduler and scheduled jobs.

Authentication: Required

cURL:

```
curl -X GET https://ottoai.shunyalabs.ai/api/v1/scheduler/status \
-H "X-API-Key: your_api_key_here"
```

Response (200 OK):

```
{
  "scheduler_running": true,
  "jobs": [
    {
      "id": "weekly_insights_generation",
      "name": "Weekly Insights Generation",
      "trigger": "cron",
      "next_run": "2026-01-13T00:00:00Z"
    }
  ]
}
```

2. Call Processing APIs

2.1 Submit Call for Processing

Endpoint: POST /api/v1/call-processing/process

Description: Submit a call for async processing. Returns a job ID for tracking.

Authentication: Required

Request Body:

```
{
  "call_id": "call_abc123",
  "company_id": "company_xyz789",
  "audio_url": "https://s3.amazonaws.com/bucket/call_abc123.mp3",
  "phone_number": "+14155551234",
  "duration": 320,
  "call_date": "2026-01-12T10:30:00Z",
  "metadata": {
    "agent_id": "agent_001",
    "campaign": "winter_2026"
  },
  "webhook_url": "https://your-app.com/webhooks/call-complete",
  "options": {
    "skip_rag_indexing": false,
    "skip_summary_generation": false,
    "priority": "normal"
  }
}
```

cURL:

```
curl -X POST https://ottoai.shunyalabs.ai/api/v1/call-processing/process \
-H "X-API-Key: your_api_key_here" \
-H "Content-Type: application/json" \
-d '{
  "call_id": "call_abc123",
  "company_id": "company_xyz789",
  "audio_url": "https://s3.amazonaws.com/bucket/call_abc123.mp3",
  "phone_number": "+14155551234",
  "duration": 320,
  "call_date": "2026-01-12T10:30:00Z",
  "metadata": {},
  "options": {
    "priority": "normal"
  }
}'
```

Response (202 Accepted):

```
{
  "job_id": "job_a1b2c3d4e5f6",
  "call_id": "call_abc123",
  "status": "queued",
  "message": "Call processing initiated successfully",
  "estimated_completion_time": null,
  "status_url": "/api/v1/call-processing/status/job_a1b2c3d4e5f6",
  "created_at": "2026-01-12T15:00:00Z"
}
```

Error Response (409 Conflict):

```
{
  "detail": "Call call_abc123 is already being processed"
}
```

2.2 Get Job Status

Endpoint: GET /api/v1/call-processing/status/{job_id}

Description: Check the processing status and progress of a submitted job.

Authentication: Required

Path Parameters:

- `job_id` (string, required): The job ID returned from the process call endpoint

cURL:

```
curl -X GET https://ottoai.shunyalabs.ai/api/v1/call-processing/status/job_a1b2c3d4e5f6 \
-H "X-API-Key: your_api_key_here"
```

Response - Processing (200 OK):

```
{
  "job_id": "job_a1b2c3d4e5f6",
  "call_id": "call_abc123",
  "status": "processing",
  "progress": {
    "percent": 45,
    "current_step": "summarizing",
    "steps_completed": ["downloading", "transcribing", "chunking"],
    "steps_remaining": ["validation", "storage"],
    "steps_failed": []
  },
  "started_at": "2026-01-12T15:00:05Z",
  "updated_at": "2026-01-12T15:02:30Z",
  "completed_at": null,
  "failed_at": null,
  "duration_seconds": null,
  "estimated_completion": null,
  "results": null,
  "metadata": null,
  "error": null,
  "retry_available": false,
  "retry_url": null
}
```

Response - Completed (200 OK):

```
{
  "job_id": "job_a1b2c3d4e5f6",
  "call_id": "call_abc123",
  "status": "completed",
  "progress": {
    "percent": 100,
    "current_step": "completed",
    "steps_completed": ["downloading", "transcribing", "chunking", "summarizing", "validation", "storage"],
    "steps_remaining": [],
    "steps_failed": []
  },
  "started_at": "2026-01-12T15:00:05Z",
  "updated_at": "2026-01-12T15:05:00Z",
  "completed_at": "2026-01-12T15:05:00Z",
  "failed_at": null,
  "duration_seconds": 295,
  "estimated_completion": null,
  "results": {
    "summary_url": "/api/v1/call-processing/summary/call_abc123",
    "chunks_url": "/api/v1/call-processing/chunks/call_abc123",
    "transcript_url": "/api/v1/call-processing/transcript/call_abc123"
  },
  "metadata": null,
  "error": null,
  "retry_available": false,
  "retry_url": null
}
```

Response - Failed (200 OK):

```
{
  "job_id": "job_a1b2c3d4e5f6",
  "call_id": "call_abc123",
  "status": "failed",
  "progress": {
    "percent": 20,
    "current_step": "transcribing",
    "steps_completed": ["downloading"],
    "steps_remaining": ["chunking", "summarizing", "validation", "storage"],
    "steps_failed": ["transcribing"]
  },
  "started_at": "2026-01-12T15:00:05Z",
  "updated_at": "2026-01-12T15:01:30Z",
  "completed_at": null,
  "failed_at": "2026-01-12T15:01:30Z",
  "duration_seconds": null,
  "estimated_completion": null,
  "results": null,
  "metadata": null,
  "error": {
    "code": "TRANSCRIPTION_FAILED",
    "message": "Audio file format not supported",
    "details": {
      "format": "unknown",
      "suggestion": "Use MP3, WAV, or M4A format"
    }
  },
  "retry_available": true,
  "retry_url": "/api/v1/call-processing/retry/job_a1b2c3d4e5f6"
}
```

Error Response (404 Not Found):

```
{
  "detail": "Job job_a1b2c3d4e5f6 not found"
}
```

2.3 Get Call Summary

Endpoint: GET /api/v1/call-processing/summary/{call_id}

Description: Retrieve the complete summary for a processed call.

Authentication: Required

Path Parameters:

- `call_id` (string, required): The call identifier

Query Parameters:

- `include_chunks` (boolean, optional): Include chunk summaries (default: false)

cURL:

```
curl -X GET "https://ottoai.shunyalabs.ai/api/v1/call-processing/summary/call_abc123" \
-H "X-API-Key: your_api_key_here"
```

cURL with chunks:

```
curl -X GET "https://ottoai.shunyalabs.ai/api/v1/call-processing/summary/call_abc123?include_chunks=true" \
-H "X-API-Key: your_api_key_here"
```

Response (200 OK):

```
{
  "call_id": "call_abc123",
```

```
"company_id": "company_xyz789",
"status": "completed",
"processed_at": "2026-01-12T15:05:00Z",
"summary": {
  "call_summary": "Homeowner inquired about solar panel installation. Has a 2000 sq ft home with south-facing roof. Budget of $30k",
  "key_points": [
    "Customer owns home (not renting)",
    "High electricity bills ($350/month)",
    "Interested in tax incentives",
    "Has HOA approval already"
  ],
  "sentiment_score": 0.75,
  "outcome_category": "appointment_booked",
  "next_steps": "Follow-up call scheduled for 2026-01-15"
},
"compliance": {
  "greeting_proper": true,
  "intro_delivered": true,
  "permission_asked": true,
  "closing_proper": true,
  "overall_score": 0.88,
  "violations": []
},
"objections": {
  "total_objections": 2,
  "objections_overcome": 1,
  "objections": [
    {
      "id": 1,
      "category_id": 3,
      "category_text": "Price Concern",
      "objection_text": "That seems really expensive",
      "overcome": true,
      "speaker_id": "home_owner",
      "timestamp": "00:02:45",
      "confidence_score": 0.92,
      "severity": "medium",
      "response_suggestions": [
        "Emphasize long-term savings",
        "Discuss financing options"
      ],
      "created_at": "2026-01-12T15:05:00Z"
    },
    {
      "id": 2,
      "category_id": 7,
      "category_text": "Timeline Concern",
      "objection_text": "How long will installation take?",
      "overcome": false,
      "speaker_id": "home_owner",
      "timestamp": "00:04:12",
      "confidence_score": 0.85,
      "severity": "low",
      "response_suggestions": [
        "Provide clear timeline",
        "Mention minimal disruption"
      ],
      "created_at": "2026-01-12T15:05:00Z"
    }
  ]
},
"qualification": {
  "bant_scores": {
    "need": 0.85,
    "budget": 0.70,
    "timeline": 0.55
```

```
    "timeline": 0.75,
    "authority": 0.90
  },
  "overall_score": 0.80,
  "qualification_status": "qualified",
  "booking_status": "booked",
  "appointment_type": "in_home_consultation",
  "appointment_date": "2026-01-15T14:00:00Z",
  "customer_details": {
    "name": "John Smith",
    "phone": "+14155551234",
    "email": "john.smith@example.com",
    "address": {
      "line1": "123 Main St",
      "city": "San Francisco",
      "state": "CA",
      "postal_code": "94102",
      "country": "US"
    }
  },
  "pain_points": [
    "High electricity costs",
    "Environmental concerns",
    "Energy independence"
  ],
  "pending_actions": [
    {
      "type": "send_proposal",
      "owner": "Sales Rep",
      "due_at": "2026-01-14T17:00:00Z",
      "raw_text": "I'll send you a detailed proposal by tomorrow",
      "confidence": 0.95,
      "contact_method": "email"
    }
  ]
}
```

Error Response (404 Not Found):

```
{
  "detail": "Summary for call call_abc123 not found"
}
```

2.4 Get Call Chunks

Endpoint: GET /api/v1/call-processing/chunks/{call_id}

Description: Get all chunk summaries for a call (used for semantic search).

Authentication: Required

Path Parameters:

- `call_id` (string, required): The call identifier

cURL:

```
curl -X GET https://ottoai.shunyalabs.ai/api/v1/call-processing/chunks/call_abc123 \
-H "X-API-Key: your_api_key_here"
```

Response (200 OK):

```

{
  "call_id": "call_abc123",
  "total_chunks": 5,
  "chunks": [
    {
      "chunk_id": "chunk_001",
      "chunk_index": 0,
      "summary": {
        "text": "Agent introduces company and asks permission to continue",
        "topics": ["greeting", "introduction"],
        "sentiment": 0.8
      },
      "milvus_id": "450123789456",
      "created_at": "2026-01-12T15:04:00Z"
    },
    {
      "chunk_id": "chunk_002",
      "chunk_index": 1,
      "summary": {
        "text": "Customer expresses interest in solar panels, mentions high electricity bills",
        "topics": ["need_identification", "pain_points"],
        "sentiment": 0.65
      },
      "milvus_id": "450123789457",
      "created_at": "2026-01-12T15:04:05Z"
    },
    {
      "chunk_id": "chunk_003",
      "chunk_index": 2,
      "summary": {
        "text": "Discussion of budget and financing options",
        "topics": ["budget", "financing"],
        "sentiment": 0.7
      },
      "milvus_id": "450123789458",
      "created_at": "2026-01-12T15:04:10Z"
    },
    {
      "chunk_id": "chunk_004",
      "chunk_index": 3,
      "summary": {
        "text": "Addressing timeline and installation concerns",
        "topics": ["timeline", "objections"],
        "sentiment": 0.75
      },
      "milvus_id": "450123789459",
      "created_at": "2026-01-12T15:04:15Z"
    },
    {
      "chunk_id": "chunk_005",
      "chunk_index": 4,
      "summary": {
        "text": "Booking appointment and confirming next steps",
        "topics": ["booking", "closing"],
        "sentiment": 0.85
      },
      "milvus_id": "450123789460",
      "created_at": "2026-01-12T15:04:20Z"
    }
  ]
}

```

Error Response (404 Not Found):


```
{
  "detail": "Chunks for call call_abc123 not found"
}
```

2.5 Retry Failed Job

Endpoint: POST /api/v1/call-processing/retry/{job_id}

Description: Retry a failed processing job.

Authentication: Required

Path Parameters:

- `job_id` (string, required): The job ID that failed

cURL:

```
curl -X POST https://ottoai.shunyalabs.ai/api/v1/call-processing/retry/job_a1b2c3d4e5f6 \
-H "X-API-Key: your_api_key_here"
```

Response (202 Accepted):

```
{
  "job_id": "job_x9y8z7w6v5u4",
  "original_job_id": "job_a1b2c3d4e5f6",
  "call_id": "call_abc123",
  "status": "queued",
  "message": "Job retry initiated",
  "retry_attempt": 2,
  "status_url": "/api/v1/call-processing/status/job_x9y8z7w6v5u4"
}
```

Error Response (400 Bad Request):

```
{
  "detail": "Job job_a1b2c3d4e5f6 cannot be retried (status: completed)"
}
```

Error Response (501 Not Implemented):

```
{
  "detail": "Retry not yet implemented - original request data not stored"
}
```

3. Insights APIs

3.1 Generate Weekly Insights

Endpoint: POST /api/v1/insights/generate

Description: Trigger weekly insights generation for companies. This is designed to be called by external cron jobs.

Authentication: Required

Request Body:

```
{
  "week_start": "2026-01-06",
  "week_end": "2026-01-12",
  "company_ids": ["company_xyz789", "company_abc123"],
  "insight_types": ["company", "customer", "objection"],
  "webhook_url": "https://your-app.com/webhooks/insights-complete",
  "options": {
    "force_regenerate": false,
    "include_inactive_customers": false
  }
}
```

cURL - Default (Last Week):

```
curl -X POST https://ottoai.shunyalabs.ai/api/v1/insights/generate \
-H "X-API-Key: your_api_key_here" \
-H "Content-Type: application/json" \
-d '{
  "insight_types": ["company", "customer", "objection"]
}'
```

cURL - Specific Week:

```
curl -X POST https://ottoai.shunyalabs.ai/api/v1/insights/generate \
-H "X-API-Key: your_api_key_here" \
-H "Content-Type: application/json" \
-d '{
  "week_start": "2026-01-06",
  "week_end": "2026-01-12",
  "company_ids": ["company_xyz789"],
  "insight_types": ["company", "customer", "objection"],
  "webhook_url": "https://your-app.com/webhooks/insights-complete",
  "options": {
    "force_regenerate": false
  }
}'
```

Response (202 Accepted):

```
{
  "job_id": "insight_job_abc123def456",
  "status": "queued",
  "message": "Insight generation initiated for week 2026-01-06 to 2026-01-12",
  "week_start": "2026-01-06",
  "week_end": "2026-01-12",
  "company_count": 2,
  "insight_types": ["company", "customer", "objection"],
  "estimated_completion_time": null,
  "estimated_duration": "5-10 minutes",
  "status_url": "/api/v1/insights/status/insight_job_abc123def456",
  "created_at": "2026-01-13T00:00:00Z",
  "queued_at": "2026-01-13T00:00:00Z"
}
```

Error Response (409 Conflict):

```
{
  "detail": "Insights for week 2026-01-06 already exist or are being generated"
}
```

3.2 Get Insight Generation Status

Endpoint: GET /api/v1/insights/status/{job_id}

Description: Check the status of insight generation job.

Authentication: Required

Path Parameters:

- `job_id` (string, required): The insight job ID

cURL:

```
curl -X GET https://ottoai.shunyalabs.ai/api/v1/insights/status/insight_job_abc123def456 \
-H "X-API-Key: your_api_key_here"
```

Response - Processing (200 OK):

```
{
  "job_id": "insight_job_abc123def456",
  "status": "processing",
  "week_start": "2026-01-06",
  "week_end": "2026-01-12",
  "progress": {
    "percent": 60,
    "current_step": "generating_customer_insights",
    "companies_processed": 3,
    "companies_total": 5,
    "insights_generated": {
      "company": 3,
      "customer": 45,
      "objection": 3
    }
  },
  "started_at": "2026-01-13T00:00:05Z",
  "updated_at": "2026-01-13T00:03:30Z",
  "completed_at": null,
  "failed_at": null,
  "duration_seconds": null,
  "estimated_completion": null,
  "results": null,
  "error": null,
  "retry_available": false,
  "retry_url": null
}
```

Response - Completed (200 OK):

```
{
  "job_id": "insight_job_abc123def456",
  "status": "completed",
  "week_start": "2026-01-06",
  "week_end": "2026-01-12",
  "progress": {
    "percent": 100,
    "current_step": "completed",
    "companies_processed": 5,
    "companies_total": 5,
    "insights_generated": {
      "company": 5,
      "customer": 87,
      "objection": 5
    }
  },
  "started_at": "2026-01-13T00:00:05Z",
  "updated_at": "2026-01-13T00:06:45Z",
  "completed_at": "2026-01-13T00:06:45Z",
  "failed_at": null,
  "duration_seconds": 400,
  "estimated_completion": null,
  "results": {
    "company_insights_url": "/api/v1/insights/company/current",
    "customer_insights_url": "/api/v1/insights/customers",
    "objection_insights_url": "/api/v1/insights/objections"
  },
  "error": null,
  "retry_available": false,
  "retry_url": null
}
```

3.3 Get Current Company Insight

Endpoint: GET `/api/v1/insights/company/{company_id}/current`

Description: Get the most recent weekly insight for a company.

Authentication: Required

Path Parameters:

- `company_id` (string, required): The company identifier

cURL:

```
curl -X GET https://ottoai.shunyalabs.ai/api/v1/insights/company/company_xyz789/current \
-H "X-API-Key: your_api_key_here"
```

Response (200 OK):

```

{
  "company_id": "company_xyz789",
  "week_start": "2026-01-06",
  "week_end": "2026-01-12",
  "data": {
    "total_calls": 145,
    "total_booked": 32,
    "booking_rate": 0.22,
    "avg_call_duration": 425.5,
    "avg_compliance_score": 0.87,
    "avg_sentiment_score": 0.74,
    "avg_qualification_score": 0.68,
    "top_performers": [
      {
        "rep_id": "rep_001",
        "rep_name": "Sarah Johnson",
        "calls": 28,
        "booked": 12,
        "booking_rate": 0.43,
        "avg_compliance": 0.94
      },
      {
        "rep_id": "rep_002",
        "rep_name": "Mike Chen",
        "calls": 25,
        "booked": 9,
        "booking_rate": 0.36,
        "avg_compliance": 0.91
      }
    ],
    "needs_coaching": [
      {
        "rep_id": "rep_005",
        "rep_name": "Tom Wilson",
        "calls": 18,
        "booked": 2,
        "booking_rate": 0.11,
        "issues": [
          "Low compliance score (0.65)",
          "Not handling objections effectively",
          "Missing qualification steps"
        ]
      }
    ]
  },
  "insight_heading": "Strong Performance Week",
  "insight": "Team achieved 22% booking rate, up 3% from last week. Top performers are consistently following SOP and handling obj",
  "recommendation_heading": "Focus on Coaching",
  "recommendation": "Schedule coaching sessions for reps below 15% booking rate. Focus on objection handling techniques and SOP co",
  "trends": {
    "booking_rate": "up",
    "calls": "up",
    "compliance": "stable",
    "sentiment": "up"
  },
  "week_over_week": {
    "booking_rate_change": 0.03,
    "calls_change": 12,
    "compliance_change": 0.01,
    "sentiment_change": 0.05
  }
},
"generated_at": "2026-01-13T00:06:00Z"
}

```

Error Response (404 Not Found):

```
{
  "detail": "No insights found for company company_xyz789"
}
```

3.4 Get Customer Insights List

Endpoint: GET /api/v1/insights/customers

Description: Get paginated list of customer insights with filtering options.

Authentication: Required

Query Parameters:

- `company_id` (string, required): Company identifier
- `week_start` (date, optional): Filter by week (format: YYYY-MM-DD)
- `status` (string, optional): Filter by customer status
- `priority` (string, optional): Filter by priority level (high, medium, low)
- `page` (integer, optional): Page number (default: 1)
- `limit` (integer, optional): Results per page (default: 50, max: 200)

cURL - Basic:

```
curl -X GET "https://ottoai.shunyalabs.ai/api/v1/insights/customers?company_id=company_xyz789" \
-H "X-API-Key: your_api_key_here"
```

cURL - With Filters:

```
curl -X GET "https://ottoai.shunyalabs.ai/api/v1/insights/customers?company_id=company_xyz789&priority=high&page=1&limit=20" \
-H "X-API-Key: your_api_key_here"
```

cURL - Specific Week:

```
curl -X GET "https://ottoai.shunyalabs.ai/api/v1/insights/customers?company_id=company_xyz789&week_start=2026-01-06&status=qualified" \
-H "X-API-Key: your_api_key_here"
```

Response (200 OK):

```

{
  "company_id": "company_xyz789",
  "week_start": "2026-01-06",
  "total_customers": 87,
  "page": 1,
  "limit": 50,
  "customers": [
    {
      "customer_id": "cust_001",
      "phone_number": "+14155551234",
      "customer_name": "John Smith",
      "data": {
        "calls_this_week": 3,
        "total_calls": 8,
        "current_status": "qualified",
        "status_changed": true,
        "sentiment_trend": "improving",
        "engagement_score": 0.85,
        "pending_actions": 2,
        "overdue_actions": 0,
        "last_call_date": "2026-01-12T14:30:00Z",
        "insight_heading": "Ready to Close",
        "insight": "Customer has progressed through qualification stages quickly. All BANT criteria met. Strong buying signals in la",
        "recommendation_heading": "Send Proposal ASAP",
        "recommendation": "Priority customer. Send detailed proposal today. Schedule in-home consultation within 48 hours. High conv",
        "priority": "high"
      }
    },
    {
      "customer_id": "cust_002",
      "phone_number": "+14155559876",
      "customer_name": "Maria Garcia",
      "data": {
        "calls_this_week": 1,
        "total_calls": 4,
        "current_status": "nurturing",
        "status_changed": false,
        "sentiment_trend": "stable",
        "engagement_score": 0.62,
        "pending_actions": 1,
        "overdue_actions": 1,
        "last_call_date": "2026-01-10T11:15:00Z",
        "insight_heading": "Needs Follow-up",
        "insight": "Customer interested but has timeline concerns. Mentioned waiting until Q2 2026. One overdue follow-up action.",
        "recommendation_heading": "Long-term Nurture",
        "recommendation": "Complete overdue follow-up. Set reminder for March 2026. Send educational content about financing options",
        "priority": "medium"
      }
    }
  ]
}

```

3.5 Get Objection Insights

Endpoint: GET /api/v1/insights/objections/{company_id}

Description: Get objection insights for a company showing common objections and handling effectiveness.

Authentication: Required

Path Parameters:

- **company_id** (string, required): Company identifier

Query Parameters:

- `week_start` (date, optional): Filter by week (format: YYYY-MM-DD)
- `category_id` (integer, optional): Filter by objection category (1-10)

cURL - All Objections:

```
curl -X GET https://ottoai.shunyalabs.ai/api/v1/insights/objections/company_xyz789 \
-H "X-API-Key: your_api_key_here"
```

cURL - Specific Week:

```
curl -X GET "https://ottoai.shunyalabs.ai/api/v1/insights/objections/company_xyz789?week_start=2026-01-06" \
-H "X-API-Key: your_api_key_here"
```

cURL - Specific Category:

```
curl -X GET "https://ottoai.shunyalabs.ai/api/v1/insights/objections/company_xyz789?category_id=3" \
-H "X-API-Key: your_api_key_here"
```

Response (200 OK):

```
{
  "company_id": "company_xyz789",
  "week_start": "2026-01-06",
  "week_end": "2026-01-12",
  "total_categories": 5,
  "objections": [
    {
      "category_id": 3,
      "category_name": "Price Concern",
      "total_occurrences": 42,
      "overcome_rate": 0.64,
      "avg_severity": 2.3,
      "trend": "stable",
      "example_objections": [
        "That's too expensive",
        "I can't afford that right now",
        "Your competitors are cheaper"
      ],
      "top_responses": [
        "Emphasize long-term ROI and savings",
        "Discuss flexible financing options",
        "Highlight unique value propositions"
      ],
      "insight_heading": "Price Remains Top Objection",
      "insight": "Price concerns appeared in 29% of calls. 64% overcome rate shows room for improvement. Top performers consistently",
      "recommendation": "Update price objection handling script. Add financing calculator to proposal. Train reps on value-based sel
    },
    {
      "category_id": 7,
      "category_name": "Timeline Concern",
      "total_occurrences": 28,
      "overcome_rate": 0.82,
      "avg_severity": 1.8,
      "trend": "improving",
      "example_objections": [
        "How long will installation take?",
        "I need it done by summer",
        "Can you start next week?"
      ],
      "top_responses": [
        "Provide clear timeline expectations",
        "Explain the installation process",
        "Offer priority scheduling options"
      ],
      "insight_heading": "Timeline Concerns Well-Handled",
      "insight": "82% overcome rate on timeline objections. Customers respond well to clear timeline communication.",
    }
  ]
}
```



```
    "recommendation": "Continue current approach. Consider creating visual timeline infographic for proposals."
  },
  {
    "category_id": 1,
    "category_name": "Need Clarification",
    "total_occurrences": 21,
    "overcome_rate": 0.90,
    "avg_severity": 1.5,
    "trend": "down",
    "example_objections": [
      "I'm not sure I need this",
      "Why should I switch now?",
      "Is it worth it?"
    ],
    "top_responses": [
      "Ask discovery questions about pain points",
      "Calculate current costs vs. future savings",
      "Share relevant case studies"
    ],
    "insight_heading": "Strong Need Identification",
    "insight": "90% overcome rate shows excellent discovery and qualification. Objections decreasing week-over-week.",
    "recommendation": "Document best practices from top performers. Use as training material for new reps."
  }
],
"generated_at": "2026-01-13T00:06:15Z"
}
```

Error Response (404 Not Found):

```
{
  "detail": "No objection insights found"
}
```

4. Ask Otto (Conversational AI) APIs

4.1 Create Conversation

Endpoint: POST /api/v1/ask-otto/conversations

Description: Create a new conversation session with Otto AI.

Authentication: Required

Request Body:

```
{
  "company_id": "company_xyz789",
  "user_id": "user_abc123",
  "metadata": {
    "source": "web_app",
    "user_role": "sales_manager"
  }
}
```

cURL:

```
curl -X POST https://ottoai.shunyalabs.ai/api/v1/ask-otto/conversations \
-H "X-API-Key: your_api_key_here" \
-H "Content-Type: application/json" \
-d '{
  "company_id": "company_xyz789",
  "user_id": "user_abc123",
  "metadata": {
    "source": "web_app"
  }
}'
```

Response (201 Created):

```
{
  "conversation_id": "conv_x1y2z3a4b5c6",
  "company_id": "company_xyz789",
  "user_id": "user_abc123",
  "created_at": "2026-01-12T15:30:00Z",
  "message_count": 0,
  "expires_at": "2026-01-19T15:30:00Z"
}
```

4.2 Send Message

Endpoint: POST /api/v1/ask-otto/conversations/{conversation_id}/messages

Description: Send a message to Otto and get an AI-powered response with context from call data.

Authentication: Required

Path Parameters:

- `conversation_id` (string, required): The conversation ID

Request Body:

```
{
  "message": "What did John Smith say about pricing in his last call?",
  "context": {
    "include_customer_context": true,
    "include_call_history": true,
    "max_rag_results": 5,
    "search_filters": {}
  },
  "options": {
    "stream": false,
    "include_sources": true,
    "suggest_follow_ups": true
  }
}
```

cURL - Simple Message:

```
curl -X POST https://ottoai.shunyalabs.ai/api/v1/ask-otto/conversations/conv_x1y2z3a4b5c6/messages \
-H "X-API-Key: your_api_key_here" \
-H "Content-Type: application/json" \
-d '{
  "message": "What did John Smith say about pricing in his last call?"
}'
```

cURL - With Custom Context:

```
curl -X POST https://ottoai.shunyalabs.ai/api/v1/ask-otto/conversations/conv_x1y2z3a4b5c6/messages \
-H "X-API-Key: your_api_key_here" \
-H "Content-Type: application/json" \
-d '{
  "message": "Show me all qualified leads from last week",
  "context": {
    "include_customer_context": true,
    "include_call_history": true,
    "max_rag_results": 10,
    "search_filters": {
      "date_range": {
        "start": "2026-01-06",
        "end": "2026-01-12"
      },
      "qualification_status": ["qualified"]
    }
  },
  "options": {
    "include_sources": true,
    "suggest_follow_ups": true
  }
}'
```

Response (200 OK):

```
{
  "conversation_id": "conv_x1y2z3a4b5c6",
  "message_id": "msg_abc123def456",
  "answer": "In John Smith's most recent call on January 12th, he expressed concern about the initial $30,000 price point, saying \"",
  "sources": [
    {
      "call_id": "call_abc123",
      "chunk_id": "chunk_003",
      "corpus_type": "chunk_summary",
      "excerpt": "Customer: That seems really expensive. Agent: I understand. Let me show you the ROI over 20 years...",
      "relevance_score": 0.94,
      "call_date": "2026-01-12T10:30:00Z",
      "metadata": {
        "speaker": "home_owner",
        "timestamp": "00:02:45"
      }
    },
    {
      "call_id": "call_abc123",
      "chunk_id": null,
      "corpus_type": "call_summary",
      "excerpt": "Price objection was raised but overcome through discussion of financing options and long-term savings.",
      "relevance_score": 0.88,
      "call_date": "2026-01-12T10:30:00Z",
      "metadata": {}
    }
  ],
  "customer_context": {
    "customer_id": "cust_001",
    "phone_number": "+14155551234",
    "name": "John Smith",
    "location": "San Francisco, CA",
    "qualification_status": "qualified",
    "total_calls": 8,
    "last_call_date": "2026-01-12T10:30:00Z",
    "last_call_outcome": "appointment_booked",
    "pending_actions": [
      {
        "type": "send_proposal",
        "due_at": "2026-01-14T17:00:00Z"
      }
    ]
  },
  "suggested_follow_ups": [
    "What other objections did John Smith raise?",
    "Show me John Smith's qualification scores",
    "When is John's appointment scheduled?"
  ],
  "metadata": {
    "tokens_used": null,
    "response_time_ms": 1250,
    "rag_results_count": 2,
    "customer_context_found": true
  },
  "created_at": "2026-01-12T15:31:00Z"
}
```

Error Response (404 Not Found):

```
{
  "detail": "Conversation not found"
}
```

4.3 Get Conversation Messages

Endpoint: GET /api/v1/ask-otto/conversations/{conversation_id}/messages

Description: Retrieve message history for a conversation.

Authentication: Required

Path Parameters:

- `conversation_id` (string, required): The conversation ID

Query Parameters:

- `limit` (integer, optional): Number of messages to return (default: 50, max: 200)
- `before` (string, optional): Message ID for pagination (returns messages before this ID)

cURL - Recent Messages:

```
curl -X GET https://ottoai.shunyalabs.ai/api/v1/ask-otto/conversations/conv_x1y2z3a4b5c6/messages \
-H "X-API-Key: your_api_key_here"
```

cURL - With Pagination:

```
curl -X GET "https://ottoai.shunyalabs.ai/api/v1/ask-otto/conversations/conv_x1y2z3a4b5c6/messages?limit=20&before=msg_xyz789" \
-H "X-API-Key: your_api_key_here"
```

Response (200 OK):

```

{
  "conversation_id": "conv_x1y2z3a4b5c6",
  "messages": [
    {
      "message_id": "msg_abc123def456",
      "role": "user",
      "content": "What did John Smith say about pricing in his last call?",
      "sources": [],
      "created_at": "2026-01-12T15:31:00Z"
    },
    {
      "message_id": "msg_abc123def457",
      "role": "assistant",
      "content": "In John Smith's most recent call on January 12th, he expressed concern about the initial $30,000 price point...",
      "sources": [
        {
          "call_id": "call_abc123",
          "chunk_id": "chunk_003",
          "corpus_type": "chunk_summary",
          "excerpt": "Customer: That seems really expensive...",
          "relevance_score": 0.94,
          "call_date": "2026-01-12T10:30:00Z",
          "metadata": {}
        }
      ],
      "created_at": "2026-01-12T15:31:01Z"
    },
    {
      "message_id": "msg_abc123def458",
      "role": "user",
      "content": "What are his next steps?",
      "sources": [],
      "created_at": "2026-01-12T15:31:30Z"
    },
    {
      "message_id": "msg_abc123def459",
      "role": "assistant",
      "content": "John Smith has an in-home consultation scheduled for January 15th at 2:00 PM. There's also a pending action to sen",
      "sources": [
        {
          "call_id": "call_abc123",
          "chunk_id": null,
          "corpus_type": "call_summary",
          "excerpt": "Appointment booked for 2026-01-15 at 14:00. Follow-up proposal needed.",
          "relevance_score": 0.96,
          "call_date": "2026-01-12T10:30:00Z",
          "metadata": {}
        }
      ],
      "created_at": "2026-01-12T15:31:31Z"
    }
  ],
  "total_messages": 4,
  "has_more": false
}

```

4.4 Get Conversation Details

Endpoint: GET /api/v1/ask-otto/conversations/{conversation_id}

Description: Get details about a specific conversation.

Authentication: Required

Path Parameters:

- `conversation_id` (string, required): The conversation ID

cURL:

```
curl -X GET https://ottoai.shunyalabs.ai/api/v1/ask-otto/conversations/conv_x1y2z3a4b5c6 \
-H "X-API-Key: your_api_key_here"
```

Response (200 OK):

```
{
  "conversation_id": "conv_x1y2z3a4b5c6",
  "company_id": "company_xyz789",
  "user_id": "user_abc123",
  "created_at": "2026-01-12T15:30:00Z",
  "updated_at": "2026-01-12T15:31:31Z",
  "message_count": 4,
  "expires_at": "2026-01-19T15:30:00Z",
  "metadata": {
    "source": "web_app",
    "user_role": "sales_manager"
  }
}
```

Error Response (404 Not Found):

```
{
  "detail": "Conversation not found"
}
```

4.5 Delete Conversation

Endpoint: DELETE `/api/v1/ask-otto/conversations/{conversation_id}`

Description: Delete a conversation and all its messages.

Authentication: Required

Path Parameters:

- `conversation_id` (string, required): The conversation ID

cURL:

```
curl -X DELETE https://ottoai.shunyalabs.ai/api/v1/ask-otto/conversations/conv_x1y2z3a4b5c6 \
-H "X-API-Key: your_api_key_here"
```

Response (204 No Content):

(Empty response body)

Error Response (404 Not Found):

```
{
  "detail": "Conversation not found"
}
```

5. SOP Document Ingestion APIs

5.1 Upload SOP Document

Endpoint: POST `/api/v1/sop/documents/upload`

Description: Upload a Standard Operating Procedure document (PDF, DOC, DOCX) for processing and metric extraction.

Authentication: Required

Content Type: multipart/form-data

Form Fields:

- `file` (file, required): The SOP document file
- `company_id` (string, required): Company identifier
- `sop_name` (string, required): Name of the SOP
- `target_role` (string, optional): Target role (null for company-wide)
- `metadata` (string, optional): JSON string with additional metadata
- `webhook_url` (string, optional): Callback URL for completion notification

cURL - Basic Upload:

```
curl -X POST https://ottoai.shunyalabs.ai/api/v1/sop/documents/upload \
-H "X-API-Key: your_api_key_here" \
-F "file=@/path/to/sales_sop.pdf" \
-F "company_id=company_xyz789" \
-F "sop_name=Sales Call Standard Operating Procedure" \
-F "target_role=sales_rep"
```

cURL - With All Options:

```
curl -X POST https://ottoai.shunyalabs.ai/api/v1/sop/documents/upload \
-H "X-API-Key: your_api_key_here" \
-F "file=@/path/to/sales_sop.pdf" \
-F "company_id=company_xyz789" \
-F "sop_name=Sales Call Standard Operating Procedure" \
-F "target_role=sales_rep" \
-F 'metadata={"version":"2.0","department":"sales","author":"John Doe"}' \
-F "webhook_url=https://your-app.com/webhooks/sop-complete"
```

cURL - Company-Wide SOP:

```
curl -X POST https://ottoai.shunyalabs.ai/api/v1/sop/documents/upload \
-H "X-API-Key: your_api_key_here" \
-F "file=@/path/to/company_sop.pdf" \
-F "company_id=company_xyz789" \
-F "sop_name=Company-Wide Communication Standards"
```

Response (202 Accepted):

```
{
  "job_id": "sop_job_a1b2c3d4e5f6",
  "status": "queued",
  "message": "SOP document processing initiated",
  "file_name": "sales_sop.pdf",
  "file_size": 2457600,
  "status_url": "/api/v1/sop/documents/status/sop_job_a1b2c3d4e5f6",
  "created_at": "2026-01-12T16:00:00Z"
}
```

Error Response (400 Bad Request - Invalid File Type):

```
{
  "detail": "Unsupported file type: image/png. Allowed: PDF, DOC, DOCX"
}
```

Error Response (413 Payload Too Large):

```
{
  "detail": "File too large: 52428800 bytes. Maximum: 52428800 bytes (50MB)"
}
```

5.2 Get SOP Processing Status

Endpoint: `GET /api/v1/sop/documents/status/{job_id}`

Description: Check the processing status of an uploaded SOP document.

Authentication: Required

Path Parameters:

- `job_id` (string, required): The job ID from upload response

cURL:

```
curl -X GET https://ottoai.shunyalabs.ai/api/v1/sop/documents/status/sop_job_a1b2c3d4e5f6 \
-H "X-API-Key: your_api_key_here"
```

Response - Processing (200 OK):

```
{
  "job_id": "sop_job_a1b2c3d4e5f6",
  "status": "processing",
  "progress": {
    "percent": 60,
    "current_step": "extracting_metrics",
    "steps_completed": ["parsing_document", "analyzing_structure"],
    "steps_remaining": ["embedding_chunks", "storage"]
  },
  "started_at": "2026-01-12T16:00:05Z",
  "updated_at": "2026-01-12T16:02:30Z",
  "completed_at": null,
  "failed_at": null,
  "duration_seconds": null,
  "results": null,
  "error": null
}
```

Response - Completed (200 OK):

```
{
  "job_id": "sop_job_a1b2c3d4e5f6",
  "status": "completed",
  "progress": {
    "percent": 100,
    "current_step": "completed",
    "steps_completed": ["parsing_document", "analyzing_structure", "extracting_metrics", "embedding_chunks", "storage"],
    "steps_remaining": []
  },
  "started_at": "2026-01-12T16:00:05Z",
  "updated_at": "2026-01-12T16:04:30Z",
  "completed_at": "2026-01-12T16:04:30Z",
  "failed_at": null,
  "duration_seconds": 265,
  "results": {
    "sop_id": "sop_xyz123abc456",
    "sop_name": "Sales Call Standard Operating Procedure",
    "sop_type": "call_script",
    "is_company_wide": false,
    "target_role": "sales_rep",
    "metrics_extracted": 12,
    "chunks_indexed": 24,
    "page_count": 15,
    "word_count": 3450
  },
  "error": null
}
```

Response - Failed (200 OK):

```
{
  "job_id": "sop_job_a1b2c3d4e5f6",
  "status": "failed",
  "progress": {
    "percent": 30,
    "current_step": "parsing_document",
    "steps_completed": [],
    "steps_remaining": ["analyzing_structure", "extracting_metrics", "embedding_chunks", "storage"]
  },
  "started_at": "2026-01-12T16:00:05Z",
  "updated_at": "2026-01-12T16:00:45Z",
  "completed_at": null,
  "failed_at": "2026-01-12T16:00:45Z",
  "duration_seconds": null,
  "results": null,
  "error": {
    "code": "DOCUMENT_PARSING_ERROR",
    "message": "Unable to extract text from PDF",
    "details": {
      "reason": "Document may be password protected or corrupted"
    }
  }
}
```

Error Response (404 Not Found):

```
{
  "detail": "Job not found"
}
```

5.3 Get Company SOP Metrics

Endpoint: `GET /api/v1/sop/metrics/{company_id}`

Description: Get all active SOP metrics for a company (role-specific and company-wide).

Authentication: Required

Path Parameters:

- `company_id` (string, required): Company identifier

Query Parameters:

- `role` (string, optional): Filter by target role
- `sop_id` (string, optional): Get specific SOP metrics

cURL - All Company SOPs:

```
curl -X GET https://ottoai.shunyalabs.ai/api/v1/sop/metrics/company_xyz789 \
-H "X-API-Key: your_api_key_here"
```

cURL - Role-Specific:

```
curl -X GET "https://ottoai.shunyalabs.ai/api/v1/sop/metrics/company_xyz789?role=sales_rep" \
-H "X-API-Key: your_api_key_here"
```

cURL - Specific SOP:

```
curl -X GET "https://ottoai.shunyalabs.ai/api/v1/sop/metrics/company_xyz789?sop_id=sop_xyz123abc456" \
-H "X-API-Key: your_api_key_here"
```

Response (200 OK):

```
{
  "company_id": "company_xyz789",
  "active_sops": [
```

```

{
  "sop_id": "sop_xyz123abc456",
  "sop_name": "Sales Call Standard Operating Procedure",
  "sop_type": "call_script",
  "target_role": "sales_rep",
  "is_company_wide": false,
  "version": "2.0",
  "total_metrics": 12,
  "metrics": [
    {
      "metric_id": "metric_001",
      "metric_name": "Proper Greeting",
      "description": "Agent provides professional greeting with company name and personal introduction",
      "evaluation_method": "binary",
      "target_value": 1.0,
      "weight": 0.08,
      "applicable_roles": ["sales_rep", "customer_service"],
      "evaluation_criteria": {
        "required_elements": ["company_name", "agent_name", "permission_to_continue"]
      }
    },
    {
      "metric_id": "metric_002",
      "metric_name": "BANT Qualification",
      "description": "Agent qualifies lead using BANT framework (Budget, Authority, Need, Timeline)",
      "evaluation_method": "scoring",
      "target_value": 0.8,
      "weight": 0.15,
      "applicable_roles": ["sales_rep"],
      "evaluation_criteria": {
        "components": ["budget", "authority", "need", "timeline"],
        "minimum_score": 0.8
      }
    },
    {
      "metric_id": "metric_003",
      "metric_name": "Objection Handling",
      "description": "Agent properly addresses customer objections using approved techniques",
      "evaluation_method": "scoring",
      "target_value": 0.75,
      "weight": 0.12,
      "applicable_roles": ["sales_rep"],
      "evaluation_criteria": {
        "acknowledge": true,
        "empathize": true,
        "respond_with_value": true
      }
    }
  ],
  "created_at": "2026-01-12T16:04:30Z",
  "status": "active"
}
],
"company_wide_sops": [
  {
    "sop_id": "sop_abc789xyz123",
    "sop_name": "Company-Wide Communication Standards",
    "sop_type": "general_sop",
    "target_role": null,
    "is_company_wide": true,
    "version": "1.0",
    "total_metrics": 5,
    "metrics": [
      {
        "metric_id": "metric_101",
        "metric_name": "Standard Greeting",
        "description": "Agent uses standard company greeting",
        "evaluation_method": "binary",
        "target_value": 1.0,
        "weight": 0.2,
        "applicable_roles": ["all"],
        "evaluation_criteria": {
          "required_elements": ["greeting"]
        }
      },
      {
        "metric_id": "metric_102",
        "metric_name": "Active Listening",
        "description": "Agent demonstrates active listening skills",
        "evaluation_method": "scoring",
        "target_value": 0.5,
        "weight": 0.3,
        "applicable_roles": ["all"],
        "evaluation_criteria": {
          "components": ["paraphrasing", "summarizing", "reflecting"],
          "minimum_score": 0.3
        }
      },
      {
        "metric_id": "metric_103",
        "metric_name": "Clear Communication",
        "description": "Agent communicates clearly and concisely",
        "evaluation_method": "scoring",
        "target_value": 0.5,
        "weight": 0.3,
        "applicable_roles": ["all"],
        "evaluation_criteria": {
          "components": ["clarity", "conciseness", "structure"],
          "minimum_score": 0.3
        }
      }
    ]
  }
]

```

```
    "metric_name": "Professional Tone",
    "description": "Maintain professional and courteous tone throughout interaction",
    "evaluation_method": "scoring",
    "target_value": 0.9,
    "weight": 0.20,
    "applicable_roles": ["all"],
    "evaluation_criteria": {
      "no_slang": true,
      "respectful_language": true,
      "positive_phrasing": true
    }
  },
  "created_at": "2026-01-10T10:00:00Z",
  "status": "active"
}
]
```

Error Response (404 Not Found):

```
{
  "detail": "SOP not found"
}
```

5.4 Get SOP Document Details

Endpoint: GET /api/v1/sop/documents/{sop_id}

Description: Get detailed information about a specific SOP document.

Authentication: Required

Path Parameters:

- `sop_id` (string, required): The SOP document identifier

cURL:

```
curl -X GET https://ottoai.shunyalabs.ai/api/v1/sop/documents/sop_xyz123abc456 \
-H "X-API-Key: your_api_key_here"
```

Response (200 OK):

```
{
  "sop_id": "sop_xyz123abc456",
  "company_id": "company_xyz789",
  "sop_name": "Sales Call Standard Operating Procedure",
  "sop_type": "call_script",
  "target_role": "sales_rep",
  "is_company_wide": false,
  "file_info": {
    "original_filename": "sales_sop.pdf",
    "file_type": "application/pdf",
    "file_size": 2457600,
    "page_count": 15,
    "word_count": 3450
  },
  "sections": [
    {
      "title": "Introduction and Greeting",
      "page_start": 1,
      "page_end": 2
    },
    {
      "title": "Discovery Questions",
      "page_start": 3,
      "page_end": 5
    },
    {
      "title": "BANT Qualification",
      "page_start": 6,
      "page_end": 8
    },
    {
      "title": "Objection Handling",
      "page_start": 9,
      "page_end": 12
    },
    {
      "title": "Closing Techniques",
      "page_start": 13,
      "page_end": 15
    }
  ],
  "metrics_summary": {
    "total_metrics": 12,
    "by_category": {
      "greeting": 2,
      "qualification": 3,
      "objection_handling": 2,
      "closing": 3,
      "compliance": 2
    }
  },
  "status": "active",
  "version": "2.0",
  "created_at": "2026-01-12T16:04:30Z",
  "updated_at": "2026-01-12T16:04:30Z"
}
```

Error Response (404 Not Found):

```
{
  "detail": "SOP document not found"
}
```

5.5 List SOP Documents

Endpoint: GET /api/v1/sop/documents

Description: Get a paginated list of SOP documents for a company.

Authentication: Required

Query Parameters:

- `company_id` (string, required): Company identifier
- `status` (string, optional): Filter by status (active, inactive, draft)
- `target_role` (string, optional): Filter by target role
- `page` (integer, optional): Page number (default: 1)
- `limit` (integer, optional): Results per page (default: 20, max: 100)

cURL - All SOPs:

```
curl -X GET "https://ottoai.shunyalabs.ai/api/v1/sop/documents?company_id=company_xyz789" \  
-H "X-API-Key: your_api_key_here"
```

cURL - Active SOPs Only:

```
curl -X GET "https://ottoai.shunyalabs.ai/api/v1/sop/documents?company_id=company_xyz789&status=active" \  
-H "X-API-Key: your_api_key_here"
```

cURL - Role-Specific with Pagination:

```
curl -X GET "https://ottoai.shunyalabs.ai/api/v1/sop/documents?company_id=company_xyz789&target_role=sales_rep&page=1&limit=10" \  
-H "X-API-Key: your_api_key_here"
```

Response (200 OK):

```
{
  "company_id": "company_xyz789",
  "total": 8,
  "page": 1,
  "limit": 20,
  "documents": [
    {
      "sop_id": "sop_xyz123abc456",
      "sop_name": "Sales Call Standard Operating Procedure",
      "sop_type": "call_script",
      "target_role": "sales_rep",
      "is_company_wide": false,
      "status": "active",
      "metrics_count": 12,
      "created_at": "2026-01-12T16:04:30Z"
    },
    {
      "sop_id": "sop_abc789xyz123",
      "sop_name": "Company-Wide Communication Standards",
      "sop_type": "general_sop",
      "target_role": null,
      "is_company_wide": true,
      "status": "active",
      "metrics_count": 5,
      "created_at": "2026-01-10T10:00:00Z"
    },
    {
      "sop_id": "sop_def456ghi789",
      "sop_name": "Customer Service Excellence Guide",
      "sop_type": "process_guide",
      "target_role": "customer_service",
      "is_company_wide": false,
      "status": "active",
      "metrics_count": 8,
      "created_at": "2026-01-08T14:30:00Z"
    },
    {
      "sop_id": "sop_ghi789jkl012",
      "sop_name": "Legacy Sales Script (Deprecated)",
      "sop_type": "call_script",
      "target_role": "sales_rep",
      "is_company_wide": false,
      "status": "inactive",
      "metrics_count": 10,
      "created_at": "2025-11-15T09:00:00Z"
    }
  ]
}
```

5.6 Update SOP Status

Endpoint: PATCH /api/v1/sop/documents/{sop_id}/status

Description: Update the status of an SOP document (activate/deactivate).

Authentication: Required

Path Parameters:

- `sop_id` (string, required): The SOP document identifier

Request Body:

```
{
  "status": "inactive",
  "reason": "Replaced by updated version 3.0"
}
```

cURL - Deactivate:

```
curl -X PATCH https://ottoai.shunyalabs.ai/api/v1/sop/documents/sop_xyz123abc456/status \
-H "X-API-Key: your_api_key_here" \
-H "Content-Type: application/json" \
-d '{
  "status": "inactive",
  "reason": "Replaced by updated version 3.0"
}'
```

cURL - Activate:

```
curl -X PATCH https://ottoai.shunyalabs.ai/api/v1/sop/documents/sop_xyz123abc456/status \
-H "X-API-Key: your_api_key_here" \
-H "Content-Type: application/json" \
-d '{
  "status": "active",
  "reason": "Approved by management"
}'
```

Response (200 OK):

```
{
  "sop_id": "sop_xyz123abc456",
  "previous_status": "active",
  "new_status": "inactive",
  "updated_at": "2026-01-12T17:00:00Z"
}
```

Error Response (404 Not Found):

```
{
  "detail": "SOP document not found"
}
```

5.7 Delete SOP Document

Endpoint: DELETE /api/v1/sop/documents/{sop_id}

Description: Permanently delete an SOP document and all associated data.

Authentication: Required

Path Parameters:

- `sop_id` (string, required): The SOP document identifier

cURL:

```
curl -X DELETE https://ottoai.shunyalabs.ai/api/v1/sop/documents/sop_xyz123abc456 \
-H "X-API-Key: your_api_key_here"
```

Response (204 No Content):

(Empty response body)

Error Response (404 Not Found):


```
{
  "detail": "SOP document not found"
}
```

Error Response Format

All error responses follow a consistent format:

Standard Error Response:

```
{
  "detail": "Error message describing what went wrong"
}
```

Common HTTP Status Codes:

- 200 OK - Request succeeded
- 201 Created - Resource created successfully
- 202 Accepted - Request accepted for async processing
- 204 No Content - Request succeeded with no response body
- 400 Bad Request - Invalid request parameters or body
- 401 Unauthorized - Missing or invalid API key
- 404 Not Found - Resource not found
- 409 Conflict - Resource already exists or conflict with current state
- 413 Payload Too Large - Request body too large
- 422 Unprocessable Entity - Request validation failed
- 500 Internal Server Error - Server error
- 501 Not Implemented - Feature not yet implemented

Rate Limiting

All API endpoints are rate-limited:

- **Default:** 100 requests per minute per API key
- **Burst:** 10 requests per second

Rate limit headers are included in all responses:

```
X-RateLimit-Limit: 100
X-RateLimit-Remaining: 95
X-RateLimit-Reset: 1673539200
```

Webhooks

For async operations (call processing, insights generation, SOP processing), you can provide a `webhook_url` to receive completion notifications.

Webhook Payload Example (Call Processing):

```
{
  "event": "call_processing.completed",
  "job_id": "job_a1b2c3d4e5f6",
  "call_id": "call_abc123",
  "status": "completed",
  "completed_at": "2026-01-12T15:05:00Z",
  "duration_seconds": 295,
  "results": {
    "summary_url": "/api/v1/call-processing/summary/call_abc123",
    "chunks_url": "/api/v1/call-processing/chunks/call_abc123",
    "transcript_url": "/api/v1/call-processing/transcript/call_abc123"
  }
}
```

Webhook Security:

- All webhook requests include X-Otto-Signature header with HMAC-SHA256 signature
- Verify signature using your API key as the secret